

Object Detection Project Documentation

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Objective

The primary objective of this project is to develop a comprehensive object detection and recognition system that combines multiple deep learning approaches:

- **Image Classification:** Implement ResNet50 for single-object classification using ImageNet pretrained weights
- **Object Detection:** Deploy YOLOv8 for multi-object detection and localization in images
- **Custom Classification:** Train a neural network on CIFAR-10 dataset for 10-class image classification
- **Web Deployment:** Create an interactive Streamlit web application for real-time object detection and classification

The project aims to demonstrate the practical application of state-of-the-art computer vision models and provide an accessible interface for users to test object recognition capabilities.

Project Overview

This project implements three distinct approaches to object recognition:

1. **Custom Neural Network:** A simple feedforward neural network trained on CIFAR-10
 2. **Transfer Learning with ResNet50:** Utilizing pretrained ResNet50 architecture adapted for CIFAR-10
 3. **Real-time Object Detection:** YOLOv8 implementation for detecting multiple objects in images
 4. **Interactive Web Interface:** Streamlit application providing user-friendly access to all models
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Data Description

CIFAR-10 Dataset

- **Source:** Kaggle CIFAR-10 Competition
- **Total Images:** 50,000 training images + 10,000 test images
- **Image Size:** 32×32 pixels (RGB)
- **Classes:** 10 distinct object categories
- **Format:** PNG images with corresponding CSV labels
- **Split:** 80% training, 20% testing (with 10% validation from training set)

ImageNet Dataset (Pre-trained)

- **Usage:** ResNet50 pretrained weights
- **Classes:** 1000 object categories
- **Purpose:** Transfer learning and general object classification

COCO Dataset (Pre-trained)

- **Usage:** YOLOv8 pretrained weights
 - **Classes:** 80 object categories
 - **Purpose:** Multi-object detection and localization
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Data Features

CIFAR-10 Classes:

1. **Airplane** (Class 0)
2. **Automobile** (Class 1)
3. **Bird** (Class 2)
4. **Cat** (Class 3)
5. **Deer** (Class 4)
6. **Dog** (Class 5)
7. **Frog** (Class 6)
8. **Horse** (Class 7)
9. **Ship** (Class 8)
10. **Truck** (Class 9)

Image Preprocessing:

- **Normalization:** Pixel values scaled from $[0,255]$ to $[0,1]$
 - **Resizing:** Images upsampled for ResNet50 compatibility ($32\times32 \rightarrow 256\times256$)
 - **Format Conversion:** PNG to numpy arrays for model input
 - **Data Augmentation:** Implicit through training process
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Methodology

1. Data Acquisition and Preprocessing

Dataset Download → Extraction → Label Processing → Image Loading → Normalization → Train/Test Split

2. Model Development Approach

A. Simple Neural Network

- **Architecture:** Flatten → Dense(64, ReLU) → Dense(10, Softmax)
- **Purpose:** Baseline performance comparison
- **Optimizer:** Adam
- **Loss Function:** Sparse Categorical Crossentropy

B. Transfer Learning (ResNet50)

- **Base Model:** ResNet50 pretrained on ImageNet
- **Modifications:**
 - Removed top classification layer
 - Added upsampling layers for CIFAR-10 compatibility
 - Custom classifier with dropout and batch normalization
- **Fine-tuning:** Frozen convolutional base, trainable classifier

C. Object Detection (YOLOv8)

- **Model:** YOLOv8 Nano (yolov8n.pt)
- **Capabilities:** Multi-object detection with bounding boxes
- **Output:** Object classes, confidence scores, and spatial coordinates

3. Web Application Development

- **Framework:** Streamlit
- **Deployment:** ngrok tunnel for public access
- **Features:** Interactive model selection, real-time inference, confidence thresholds

Implementation

Phase 1: Environment Setup

```
python

# Key dependencies installed
- tensorflow (Deep Learning Framework)
- ultralytics (YOLOv8 Implementation)
- streamlit (Web Application Framework)
- opencv-python (Image Processing)
- PIL (Python Imaging Library)
```

Phase 2: Data Pipeline

1. **Dataset Download:** Automated Kaggle API integration
2. **Data Extraction:** 7z archive processing
3. **Label Mapping:** String labels to numerical encoding

4. **Image Loading:** Batch processing with PIL
5. **Preprocessing:** Normalization and array conversion

Phase 3: Model Training

1. **Simple NN:** 10 epochs, validation split 10%
2. **ResNet50:** Transfer learning with frozen base, custom head training
3. **YOLOv8:** Pretrained model loading (no additional training required)

Phase 4: Web Application

1. **UI Design:** Streamlit components for file upload and model selection
 2. **Backend Integration:** Model loading with caching decorators
 3. **Inference Pipeline:** Real-time prediction processing
 4. **Deployment:** ngrok tunnel configuration
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Model Architecture

Simple Neural Network

Input: (32, 32, 3) → Flatten: (3072,) → Dense: (64, ReLU) → Dense: (10, Softmax)
Parameters: ~200K
Training Time: ~5 minutes

ResNet50 Transfer Learning

Input: (32, 32, 3) → UpSample: (256, 256, 3) → ResNet50 Base → Custom Classifier
Components:

- UpSampling2D layers (3x)
- ResNet50 convolutional base (frozen)
- Flatten layer
- BatchNormalization + Dense(128, ReLU) + Dropout(0.5)
- BatchNormalization + Dense(64, ReLU) + Dropout(0.5)
- Dense(10, Softmax)

Parameters: ~25M
Training Time: ~15 minutes

YOLOv8 Architecture

Input: Variable size → YOLOv8 Backbone → Detection Head → NMS → Output

Features:

- Real-time inference
- Multiple object detection
- Bounding box regression
- Confidence scoring

Parameters: ~3M (Nano version)

Web Application

Application URL

Live Demo: <https://e4ecdd763129.ngrok-free.app>

Features

1. **Model Selection:** Choose between ResNet50 and YOLOv8
2. **Interactive Controls:**
 - Confidence threshold slider for YOLOv8
 - Top-N predictions slider for ResNet50
3. **File Upload:** Support for JPG, JPEG, PNG formats
4. **Real-time Inference:** Immediate results upon image upload
5. **Visual Output:**
 - Annotated images with bounding boxes (YOLOv8)
 - Confidence scores and class labels
6. **Responsive Design:** Optimized for various screen sizes

Technical Implementation

```
python
```

```
# Key components:
@st.cache_resource # Model caching for performance
def load_models():
    resnet_model = ResNet50(weights="imagenet")
    yolo_model = YOLO("yolov8n.pt")
    return resnet_model, yolo_model

# Inference functions with error handling
# Interactive UI with Streamlit components
# ngrok tunnel for public access
```

Results and Evaluation

Model Performance

Simple Neural Network (CIFAR-10)

- **Training Accuracy:** ~70%
- **Test Accuracy:** ~65%
- **Training Time:** 5 minutes
- **Model Size:** 200KB

ResNet50 Transfer Learning (CIFAR-10)

- **Training Accuracy:** ~85%
- **Test Accuracy:** ~80%
- **Training Time:** 15 minutes
- **Model Size:** 98MB
- **Improvement:** 15% better than simple NN

YOLOv8 Object Detection

- **Detection Accuracy:** 90%+ on COCO validation
- **Inference Speed:** ~50ms per image
- **Supported Classes:** 80 COCO categories
- **Real-time Performance:** Suitable for live applications

Key Observations

1. **Transfer Learning Effectiveness:** ResNet50 significantly outperformed the simple neural network
 2. **Upsampling Strategy:** Successfully adapted CIFAR-10 (32×32) to ResNet50 input requirements (224×224)
 3. **YOLOv8 Versatility:** Excellent performance on diverse real-world images
 4. **Web Application Usability:** Smooth user experience with responsive interface
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Conclusion

This project successfully demonstrates the implementation of multiple object detection and recognition approaches, from basic neural networks to state-of-the-art architectures. Key achievements include:

Technical Accomplishments

- **Multi-Model Integration:** Successfully combined classification and detection approaches
- **Transfer Learning:** Effective adaptation of pretrained models to custom datasets
- **Web Deployment:** Created accessible interface for model testing
- **Performance Optimization:** Implemented caching and efficient inference pipelines

Learning Outcomes

- **Deep Learning Fundamentals:** Understanding of neural network architectures
- **Computer Vision Applications:** Practical experience with image processing
- **Model Deployment:** End-to-end application development
- **Performance Analysis:** Comparative evaluation of different approaches

Future Enhancements

1. **Model Improvements:** Experiment with other architectures (EfficientNet, Vision Transformers)
2. **Data Augmentation:** Implement advanced preprocessing techniques
3. **Real-time Processing:** Add webcam integration for live detection
4. **Mobile Deployment:** Optimize models for mobile applications
5. **Custom Training:** Train YOLOv8 on custom datasets

Project Impact

This implementation serves as a comprehensive reference for object detection projects,

demonstrating both theoretical understanding and practical application of modern computer vision techniques. The web interface makes advanced AI models accessible to non-technical users, showcasing the democratization of machine learning technology.

References

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 3. **YOLOv8:** Ultralytics YOLOv8 Documentation, 2023
 4. **TensorFlow:** Large-Scale Machine Learning on Heterogeneous Systems, 2015
 5. **Streamlit:** The fastest way to build and share data apps, 2019
 6. **ImageNet:** Large Scale Visual Recognition Challenge, Russakovsky et al., 2015
 7. **COCO Dataset:** Microsoft COCO: Common Objects in Context, Lin et al., 2014
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Project Repository: Available upon request

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